NTNU

Kunnskap for en bedre verden

Department of Ocean Operations and Civil Engineering

MMA4002 Design for Automated Manufacturing

Assignment 3 - Lean

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1 Value Stream Map(s)

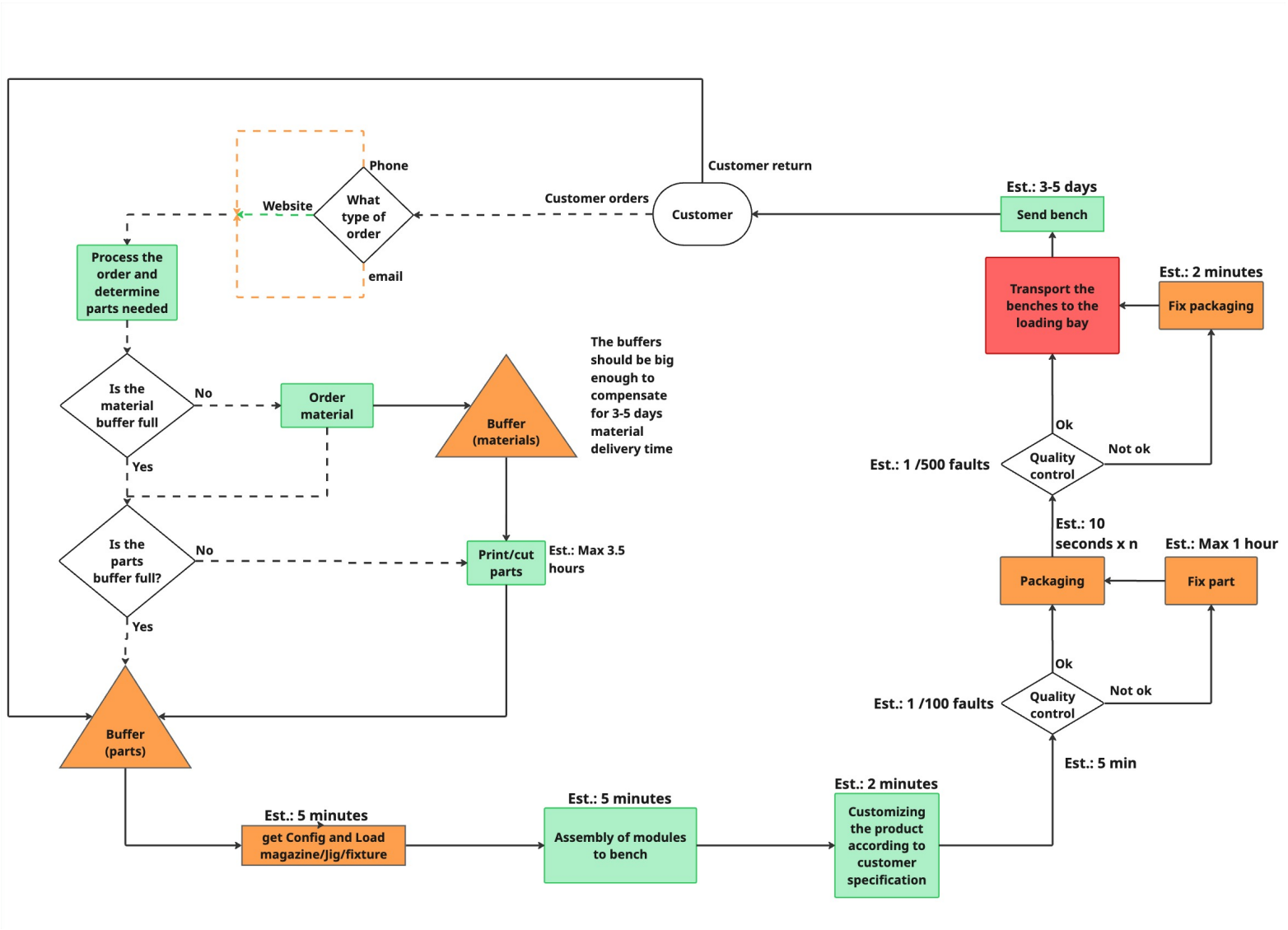


Figure 1: Caption

1.1 Explanation of the nodes

Different types of orders: In this process, phone, email and website orderings has been included. This is because different customers may have different needs.

Processing the order: Processing of the order has also been included. The intent is that a computer program automatically verifies the orders and determines what parts are needed to fulfill the order.

Buffers and ordering of materials: The buffers are set as triangles, but should not be thought of strictly as a warehouse. The idea is that the buffers should be large enough to compensate for delivery times from the suppliers with an additional small margin of safety. This ensures that the factory can continue production at full speed, if desired, while waiting for deliveries from the suppliers. This way, the company avoids tying capital in a giant warehouse full of spares that would be wasteful (Muda) most of the time.

If the buffers are not full, new materials will be ordered as soon as possible (or instantly if done automatically).

Loading the magazines, jigs and fixtures: Parts are loaded into feeder magazines, jigs and fixtures automatically. In the ManuLab, this will probably be done either manually or by a mobile robot. In a real factory however, this would probably be done with a (short) conveyor belt or similar.

The assembly: The assembly is pretty self-explanatory, but in short, this process entails mounting the planks to the bench-frames in the jigs/fixtures.

Customization of the product: This step enables the products to be customized with different colors, engraving of the name plates or logos or similar.

Quality control and fixing the product if necessary: A quality control step has been added to ensure the product live up to laws, regulations and the predefined standards of the product. This could be done with visual inspections (either by humans or machine vision), standardized automatic testing of loads and vibrations and similar. If the product does not meet the expectations, the product may be fixed by humans. If a QC fails the product gets put aside to be fixed so it doesn't add wait step to the cycle time.

Packaging and quality control of the packaging: The benches will be stacked and packaged together in a packaging machine. The quality control of the packaging is likely a bit overkill, but could be worth it if done automatically.

Transporting the benches: The benches are then transported from the robot cells to the loading bay/dock. In the manulab, this is performed with mobile robots.

2 Analysis

2.1 Coloring of nodes and improvements

Order type:
Website orders has been marked green, while phone and email orders are marked as orange (necessary waste) because phone and email orders take more human resources than electronic orders which can be almost entirely automated.

Processing the order:
Processing the order is an automated (almost) instantaneous process which adds value by predetermining what parts are needed to avoid waste further down the line.

Ordering materials and printing/cutting:
Ordering materials, printing and cutting is also set to green (value adding).

Buffers:
The buffers are set to orange. This is because buffers tie up capital, which is a waste, while also providing a safety net to prevent a full shutdown in the case of materials or parts not arriving in time. Therefore, they are deemed both wasteful and necessary.

Preparing for robot assembly:
To prepare for the robot assembly in the robot cell, the parts has to be loaded into the magazines. This is set to orange, necessary waste, because it doesn't really provide any value to the customer. It is however necessary in order to actually produce the product.

Assembly:
The assembly is set to green because it creates value for the customer.

Customization:
The customization of the product is also set to green because it creates value for the customer. If the customer doesn't want any customization, the step is simply omitted and therefore doesn't create any waste for the customers that don't want it.

Quality control:
One of the more wasteful steps in the current process flow map is probably the quality control steps. While quality control may be necessary in some cases, it can be very wasteful in other. The way to effectivize quality control, according to the lean principle of Jidoka, is to continuously automatically test the products with machines to see if they meet the requirements and stop the line if they don't. Then one could investigate the problem, fix the problem at the source and prevent similar problems from happening again. This creates more robust processes and products, less resources to fix failed products and gives a better flow compared to failed

products piling up in a warehouse to be fixed.

If one irons out these problems over time, one could, in theory, abandon the quality control steps entirely and leave the ultimate quality control up to the machines in the factory and the customers instead.

Fixing the failed products:
Fixing failed products is also set to orange for the same reason as mentioned above: In the current process it is wasteful yet necessary.

Packaging:
Packaging is set to orange because it is a wasteful process, but necessary in order to protect the products during delivery.

Quality control of and fixing the packaging:
Checking the packaging and fixing it could be valuable for avoiding damages during delivery and reducing customer returns. Then it becomes necessary waste.

However, one could also argue that this step is unnecessary waste. Another, probably smarter, way to do it, however, is probably to make sure the packaging-machines do it correctly the first time (in accordance to Jidoka).

Transport:
The transport using mobile robots is set to red, unnecessary waste. The transport inside the factory does not create any value for the customer and is therefore wasteful. Some transport may be necessary in some cases, but this should be limited to the absolute minimum possible.

2.2 Timeline

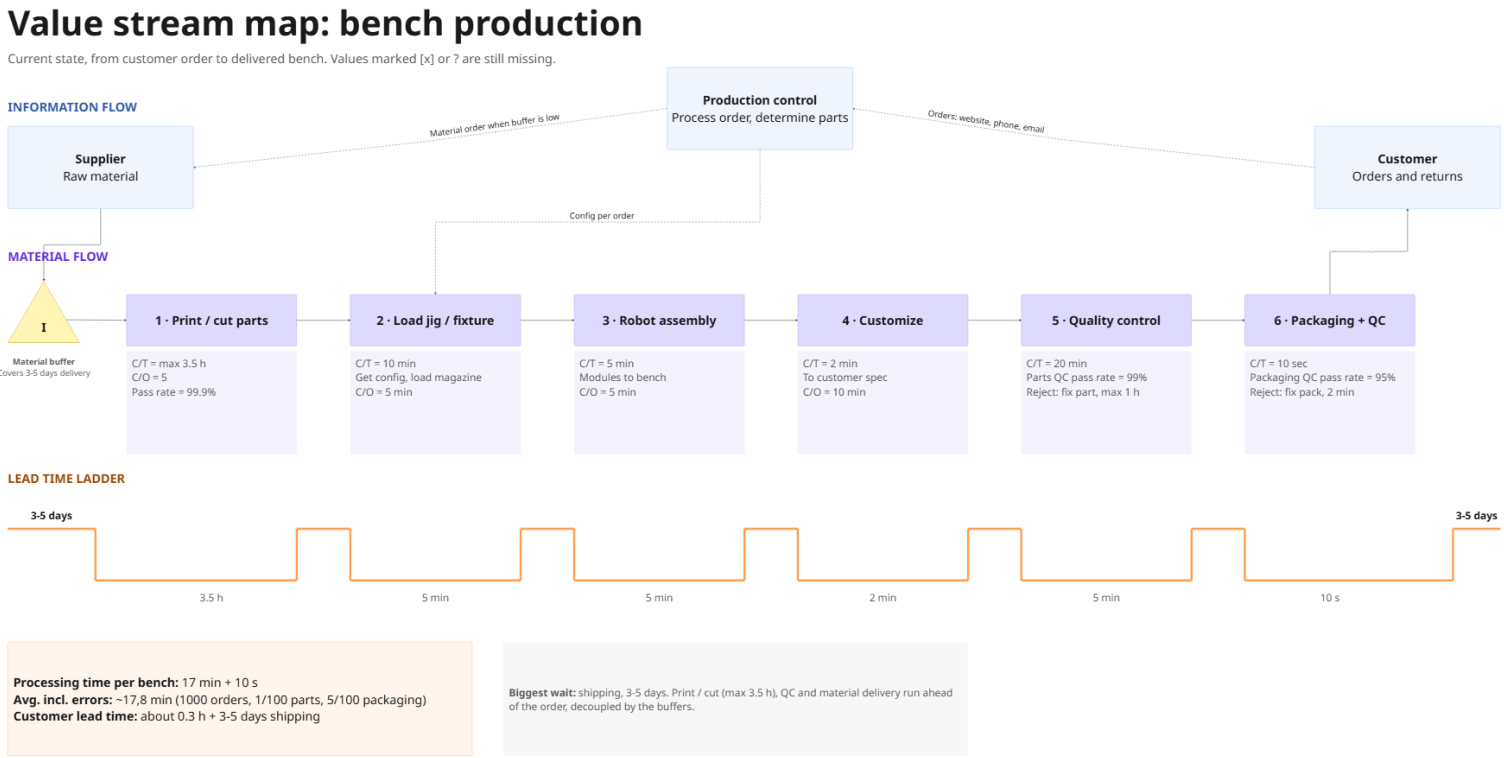


Figure 2: Diagram created with help from Claude. It sued our data and created the necessary boxes to make a VSM and timeline

The customer demand is $D = 1000$ benches over a six-month period. Production runs in shifts all year round, and holidays are planned around customer orders.

All process times below are **estimates**, not measured data.

$$t_{\text{day}} = 24\,\text{h} - \underbrace{0.5\,\text{h}}_{\text{setup}} - \underbrace{0.5\,\text{h}}_{\text{stops}} - \underbrace{4 \times 0.5\,\text{h}}_{\text{breaks}} = 20\,\text{h}$$

$$T_{\text{avail}} = t_{\text{day}} \times 30.4\,\frac{\text{days}}{\text{month}} \times 6\,\text{months} = 20 \times 30.4 \times 6 = 3648\,\text{h}$$

$$T_{\text{takt}} = \frac{T_{\text{avail}}}{D} = \frac{3648\,\text{h}}{1000\ \text{benches}} = 3.648\,\frac{\text{h}}{\text{bench}} \approx 13\,133\,\text{s}$$

Process step (estimate)	Time	Class
Transport from buffer	5 min	NNVA
Assembly	5 min	VA
Custom job	2 min	VA
Quality control (QC)	5 min	NNVA
Packaging	10 s	NNVA
Fault-free cycle t_c	1030 s	

Table 1: Estimated cycle time per bench. VA = value-adding, NNVA = necessary non-value-adding.

Process cycle time

$$t_c = (5 + 5 + 2 + 5)\,\text{min} + 10\,\text{s} = 1020\,\text{s} + 10\,\text{s} = 1030\,\text{s}$$

Rework is modelled with estimated error rates: $Q_r = 1/100$ of parts need repair ($t_Q = 60\,\text{min}$ each), and $P_r = 5/100$ need repackaging ($t_P = 2\,\text{min}$ each). For a batch of n benches:

$$T_{\text{batch}} = \underbrace{n\,t_c}_{\text{fault-free}} + \underbrace{n\,Q_r\,t_Q}_{\text{part repair}} + \underbrace{n\,P_r\,t_P}_{\text{repackaging}}$$

Takt time vs. cycle time ($n = 1000$)

$$\begin{aligned}
n\,t_c &= 1000 \times 1030\,\text{s} = 1\,030\,000\,\text{s} = 286.1\,\text{h} \\
T_{\text{rework}} &= \underbrace{1000 \cdot \tfrac{1}{100} \cdot 60\,\text{min}}_{600\,\text{min}} + \underbrace{1000 \cdot \tfrac{5}{100} \cdot 2\,\text{min}}_{100\,\text{min}} = 700\,\text{min} = 11.7\,\text{h} \\
T_{\text{batch}} &= 286.1\,\text{h} + 11.7\,\text{h} = 297.8\,\text{h} \\
\bar{t}_c &= \frac{T_{\text{batch}}}{n} = \frac{297.8\,\text{h}}{1000} = 0.2978\,\text{h} \approx 17.9\,\text{min}
\end{aligned}$$

The takt time is 3.648 h per bench, but the average cycle time is only 0.298 h, so $T_{\text{takt}}/\bar{t}_c \approx 12$. All 1000 benches take 297.8 h of the 3648 h available ($\approx 8\,\%$). The cell therefore has large spare capacity, and more orders are needed to use it.

Buffer size

With a maximum buffer of 3 production days:

$$\frac{3 \times 20\,\text{h}}{\bar{t}_c} = \frac{60\,\text{h}}{0.2978\,\text{h}} \approx 201.5 \;\Rightarrow\; 201 \text{ benches of material}$$

Appendix